

- basis: if $\{x_i\}$ is a basis for V then each $v \in V$ can be represented as a linear combination of $\{x_i\}$ in exactly one way

- dimension: size of any basis
ex: $\mathbb{C}$ as a vector space over $\mathbb{R}$, $\{i, 1\}$ or $\{-i, -1\}$
or $\{1+i, 1-i\}$

- not all vector spaces have finite dimension
ex: differentiable functions

Note: we have been talking abt bases as sets (unordered, no multiplicity) going forward we'll talk abt a list of vectors forming a basis
eg. $[1, 2] \neq [2, 1]$
 $[1, 1] \neq [1]$
(ordered, allows multiplicities)

Consider 2 bases for $\mathbb{R}^2$

$$B_1 := [(1, 0), (0, 1)], \quad B_2 := [(0, 1), (1, 0)]$$

$$(2025, 3\pi) \in \mathbb{R}^2 \text{ relative to } B_1: (2025, 3\pi) = 2025(1, 0) + 3\pi(0, 1)$$

$$\text{relative to } B_2: (2025, 3\pi) = 3\pi(0, 1) + 2025(1, 0)$$

in general, if $B = [b_1, \dots, b_n]$ is a basis for V then for each $v \in V$ there are unique $v_i \in F$ s.t. $v = v_1 b_1 + \dots + v_n b_n$

The list $[v_1, \dots, v_n]$ is referred to as the coefficients of coordinates of v relative to B

$$[3\pi, 2025] \text{ coordinates relative to } B_2 \quad [2025, 3\pi] \text{ coordinates relative to } B_1$$

if I'm thinking of B_1 for $\mathbb{R}^2$ and you're thinking of B_2 for $\mathbb{R}^2$ then $[2025, 3\pi]$ means dif. vectors for you and me

relative to B_1 , $[2029, 3\pi] = 2029(1, 0) + 3\pi(0, 1) = (2029, 3\pi)$

relative to B_2 , $[2029, 3\pi] = 2029(0, 1) + 3\pi(1, 0) = (3\pi, 2029)$

all to say agreement abt bases matters when using coefficients.

Notation: if we fix a basis B for V , we often write the list of coefficients for vectors in V vertically

ex: $[2029, 3\pi] = \begin{pmatrix} 2029 \\ 3\pi \end{pmatrix}$

Ex: $P_1 = \{ax + b \mid a, b \in \mathbb{R}\}$ $B = [x, 1]$

What is the coordinate representation of $2029x + 3\pi$ relative to B ?